

Log returns are also called *growth rates*, and growth occurs *exponentially*. That is, growth will eventually cause a portfolio to increase to infinity or decrease to zero in the long run.

The log return corresponding to an arithmetic return of 10% is $\bar{r}_{t+1} = \ln(1.10) = 9.53\%$. The arithmetic return of $r_{t+1} = 10\%$ and the log return of $\bar{r}_{t+1} = 9.53\%$ are both equivalent because they represent the same end-of-period wealth of \$1.10, or gross return of $R_{t+1} = 1.10$, for every \$1 invested at the beginning of the period. With the arithmetic return, the interest is returned only at the end of the period. In contrast, the log return assumes that interest is earned at every instant (hence returns are said to be “continuously compounded”). Since interest compounds on the interest at every moment in time, the continuously compounded interest rate does not need to be as high—it is only 9.53%—to reach the same end of period wealth of 1.10 at the end of the period.

Log returns aggregate over time. If we have an asset earning gross returns of $R_{t+1} = 1.10$ for three years, we can sum three years of log returns of $\bar{r}_{t+1} = 9.53\%$ to obtain the cumulated wealth at the end of three years:

$$\begin{aligned} W_{t+3} &= \exp(\bar{r}_{t+1}) \exp(\bar{r}_{t+2}) \exp(\bar{r}_{t+3}) \\ &= \exp(\bar{r}_{t+1} + \bar{r}_{t+2} + \bar{r}_{t+3}) \\ &= \exp(3 \times 0.0953) = 1.3310 \end{aligned}$$

Log returns do not aggregate across assets. Suppose we take asset A with a gross return of $R_{t+1}^A = 1.10$ and asset B with a gross return of $R_{t+1}^B = 1.05$. These assets have log returns of $\bar{r}_{t+1}^A = \ln(1.10) = 9.53\%$ and $\bar{r}_{t+1}^B = \ln(1.05) = 4.88\%$, respectively. If we hold a portfolio of 50% asset A and 50% asset B, then the log return of this portfolio is

$$\begin{aligned} r_{t+1}^p &= \ln(0.5 \exp(\bar{r}_{t+1}^A) + 0.5 \exp(\bar{r}_{t+1}^B)) \\ &= \ln(0.5 \exp(0.0953) + 0.5 \exp(0.0448)) = \ln(1.075) = 7.23\% \\ &\neq 0.5 \times \bar{r}_{t+1}^A + 0.5 \times \bar{r}_{t+1}^B. \end{aligned}$$

That is, the log return of the portfolio is not the weighted average of each stock's log return.

1.4. ARITHMETIC VERSUS LOG RETURNS

Many practitioners and academics have long argued about whether it is better to use arithmetic or log returns.¹ Whether you use arithmetic or log returns is like

¹The debate continues today. Advocates of the arithmetic mean typically like the fact that it is unbiased. Those who prefer log returns note that they measure the actual change in wealth, and they